

PROJECT DESIGN PHASE-II

Solution Requirements Specification (SRS)

Date	18 June 2026	Project Name	ShopEz (MERN Full Stack)
Team ID	--	Maximum Marks	4 Marks
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Functional Requirements (FR)

ID	Requirement (Epic)	Description & Logic
FR-1	Member On-boarding • Data Entry • Input Validation	Provides a functional pathway for new users to register an account with their email while ensuring passwords are securely processed via cryptographic hashing.
FR-2	Session Authentication • Secure Token Handling • Auth Lifecycle	A robust sign-in mechanism that utilizes JSON Web Tokens (JWT) to maintain stateless user sessions across the storefront.
FR-3	Dynamic Storefront • Real-time Sync • Tiled Interface	Displays product inventory segmented by Men, Women, and Kids, fetching synchronized assets directly from the MongoDB cluster.
FR-4	Intelligent Lookup • Keyword Analysis	Enables customers to efficiently query the product catalog using text-based keywords with near-instantaneous search result population.
FR-5	UI-Driven Filtering • Price Range • Category Mapping	Implements live filtering of the product grid based on price, department, or size without requiring browser refreshes, handled via React state.
FR-6	Persistent Basket • Inventory CRUD • State Sync	Allows shoppers to manage their selected items—including additions, removals, and quantity adjustments—within a persistent shopping cart environment.
FR-7	Checkout Workflow • Logistics Detail • Pricing Review	A multi-step process to collect shipping information and allow final order verification before committing to a transaction.
FR-8	Logistics Visibility • Shipment Lifecycle	A user-facing portal that tracks current order milestones, from initial pending status to final shipment and delivery.
FR-9	Admin Inventory Control • Media Upload • Pricing Logic	A secure, permission-based dashboard for store managers to add, modify, or archive items within the inventory database.
FR-10	Order Oversight • Fulfillment Ops • Analytics	Centralized administrative tools to oversee global transactions, update order statuses, and evaluate high-level sales performance.

Non-Functional Requirements (NFR)

ID	Attribute	Requirement Specification	Importance
NFR-1	Security	Encryption of user passwords using Bcrypt; all backend API pathways must be fortified with JWT-based verification.	CRITICAL
NFR-2	Performance	Maximum initial page loading time of < 2 seconds; API response latency for catalog queries must remain under 500ms.	HIGH
NFR-3	Scalability	The system framework must support horizontal growth through MongoDB Atlas sharding and partition management.	HIGH
NFR-4	Responsiveness	Utilization of an adaptive, mobile-primary design strategy to ensure seamless usability across various device form factors.	HIGH
NFR-5	Availability	Targets a system uptime of 99.9% to maintain platform access even during peak commercial demand and high traffic.	CRITICAL
NFR-6	Data Integrity	Ensures complete referential consistency across the database collections for Users, Inventory, and Order history.	CRITICAL